
Correct Scores, Wrong Scientific Conclusions: Auditing AI Forecasts of Chaotic Dynamics

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Abstract

Forecast accuracy does not by itself establish scientific forecasting ability. We test five conditions that connect a score to a scientific claim: held-out generalization, complete failure accounting, matched disclosure, structural identifiability, and cross-system transfer. We apply these tests to k -pendulum ($k \in \{1, 2, 3\}$) and Lorenz-63 forecasts from language models, a time-series foundation model, learned dynamics, sparse identification, and numerical integrators. Each test changes a substantive conclusion. Learned-model error rises from 0.09–0.63 rad on training trajectories to 0.97–1.14 rad on held-out trajectories, where linearized dynamics and nearest-neighbor lookup perform better. Retaining failed forecasts lowers the reported performance of models with incomplete coverage. An unmatched comparison says that hiding physical constants helps; a paired intervention instead finds a small, model-dependent effect (pooled +0.069 rad; +0.211 to −0.042 across models). Two exact scale symmetries prove that the absolute constants are not identifiable from angle trajectories. Increasing the Neural ODE training set improves error from 1.090 to 0.861 rad but does not reach known-equation integration. The pendulum ordering also fails to transfer to Lorenz-63, where SINDy recovers the polynomial dynamics to numerical precision. On video-tracked hardware the roles invert: the RK4 model with published constants is itself imperfect, and its lead over simple baselines is gone within a second. These results show that a correct leaderboard can still support incorrect scientific conclusions: an unaudited score does not earn trust. We release the prompts, outputs, checkpoints, seeds, and compute records needed to repeat the audit.

1 Introduction

Scientific claims require more than low forecast error. A leaderboard states what was measured, not what inference it supports: a result may depend on training trajectories, dropped failures, unmatched inputs, an unidentifiable target, or one favorable system. We study the evaluation protocol itself, not only the models it ranks. Given state $\mathbf{s}_t$, the task is to predict $\mathbf{s}_{t+T}$. The k -link pendulum is integrable at $k=1$ and chaotic for $k \geq 2$, has high-precision ground truth, and permits controlled changes to physical metadata. Lorenz-63 provides a second system with a different observation model and hypothesis class; a hardware replication of the audit (§5.5) scores the reference model itself as the imperfect one.

Our contribution is a claim audit for dynamical-system forecasts. It scores only held-out trajectories, counts every failed forecast, toggles disclosure on identical trajectories, checks identifiability before scoring parameter recovery, and tests whether conclusions transfer to a second system; throughout, it records the information each method receives, without which none of the five is interpretable. The five are indexed to the claims, not drawn from a taxonomy: each tests one inference step between a score and a conclusion (sample to population, answered cells to full task, correlation to attribution,

Table 1: Conclusions before and after the claim audit. Each row changes one part of the evaluation while preserving the relevant model outputs or comparison set.

Test	Unaudited conclusion	Audited conclusion
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Generalization	Learned dynamics forecast accurately.	At this budget, held-out error is up to $11\times$ in-sample, and simple baselines perform better.
Disclosure	Withholding constants improves forecasts.	Matched, the effect is small and model-dependent ($+0.069$ rad pooled).
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

accuracy to parameter recovery, one system to a family claim), and claims we do not make would need other checks, such as calibration for probabilistic forecasts. Table 1 states the conclusion before and after each test. Empirically, all five tests alter or reject a conclusion supported by the unaudited comparison, and at least two of them concern conclusions drawn in practice: zero-shot forecasters are routinely scored without failure accounting or matched controls [Gruver et al., 2023, Zhang and Gilpin, 2025].

2 Related work

Learned dynamics and foundation forecasters. Neural ODEs learn a continuous-time vector field [Chen et al., 2018]; Hamiltonian and Lagrangian Neural Networks add energy-conserving and Euler-Lagrange inductive biases [Greydanus et al., 2019, Cranmer et al., 2020]; chaotic systems are standard testbeds [Gilpin, 2021]. LLMs act as zero-shot forecasters when series are serialized as text [Gruver et al., 2023], time-series foundation models such as Chronos, TimesFM, and Moirai are trained for general forecasting [Ansari et al., 2024, Das et al., 2024, Woo et al., 2024], and Zhang and Gilpin [2025] apply them to chaotic systems. LLMs also do well on mathematical [Lewkowycz et al., 2022, Wei et al., 2022] and physics [Xu et al., 2025] reasoning. We evaluate all of these on one grid.

Evaluation validity. Our five checks are the standard validity questions of measurement—internal (do failure accounting and matched controls isolate the effect), construct (does the score measure forecasting rather than input access or an unidentifiable target), and external (does a conclusion transfer). SINDy [Brunton et al., 2016] and NVAR [Gauthier et al., 2021] supply an identification baseline; structural identifiability asks which parameters are recoverable from observed outputs, including under symmetries [Borgqvist et al., 2024]; and unlike selective prediction, which reports risk with coverage [Geifman and El-Yaniv, 2017], we keep failed cells in the score at fixed coverage.

3 Benchmark design

Choice of systems. We use the k -link pendulum to vary the number of coupled links while retaining a high-precision numerical solution. Its constants can also be hidden without changing the underlying trajectory, which permits the matched test in §5.2. The smaller Lorenz-63 experiment repeats the local-model comparison on a second chaotic system; the LLM and disclosure experiments were not rerun there.

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to predict $\mathbf{s}_T$ at horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4 at this step, the RK4 baseline reported later (the same integrator at a $10\times$ coarser $\Delta t=10^{-2}$) is a numerical reference by construction, not an independent competitor. Euler and the symplectic integrator are distinct known-physics methods. Inputs available to each family are stated in §3.3.

Table 2: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos-2 is history-conditioned; learned hidden cells withhold hidden constants.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos-2)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	✓ except <code>changed_hidden</code>	—

75 3.2 Regimes and grid

76 Three regimes test generalization to non-standard constants (Table 9). In `changed_hidden` the
77 constants (g, L, m, d), with damping d , are not disclosed to the model. This regime contrast is not
78 matched—`changed_hidden` differs in its constants as well as in disclosure, so its trajectories differ
79 in difficulty—which is why §5.2 adds a disclosure toggle on identical trajectories. Withholding is
80 nonetheless informative, since a single state contains no observed dynamics from which to identify the
81 constants. An evaluation cell is a tuple (model, k , regime, horizon), with horizons $\{0.01, 1, 10, 60\}$ s.
82 The full LLM grid also varies input modality and prompting; every comparison here uses the
83 coordinate modality with direct prompting (the other axes are analyzed in Appendix A).

84 **Out-of-sample protocol.** Learned dynamics models are never scored on their training data. We
85 generate a shared held-out set (a separate seed from the training trajectories) of 180 initial conditions
86 (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate every family on the identical trajectories, keyed
87 by `movement_id`. The cross-family comparison (§5.1) uses horizons $\{1, 10\}$ s, where chaos has
88 begun but error has not saturated (§3.4), giving $180 \times 2 = 360$ cells per model; in-sample error
89 appears only once, to quantify the size of the generalization gap (§5.1).

90 3.3 Inputs and grouping

91 Table 2 states what each family receives. Chronos is a time-series model, so we evaluate it in its
92 standard form: a history-conditioned forecaster with a 10 s pre- $t=0$ context window. The LLMs and
93 learned dynamics models are state-conditioned from $t=0$. For learned models, hidden-regime cells
94 are equalized by withholding hidden constants and using normal-regime prior constants. A separate
95 asymmetry runs in the opposite direction: the LLMs are large hosted models, whereas the learned
96 models are small (256-wide, 3-layer MLPs trained on a single 8 GB GPU; Appendix B). The behavior
97 of larger physics-informed networks is outside this study’s scope. We separate known-physics
98 methods (RK4, Euler, symplectic) from black-box methods (LLMs, Chronos, learned dynamics); the
99 former are numerical references.

100 3.4 Metrics

101 The primary metric is mean absolute angle error (rad), wrapped to $[-\pi, \pi]$ and averaged over links
102 and trajectories; energy drift is reported separately (§5.6).

103 **Reliability-adjusted error.** A cell is answered only if the model returns a parseable, finite prediction.
104 Reporting error over answered cells alone would reward abstention on hard cells. We therefore assign
105 unanswered cells the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform
106 guess); under the strict upper bound π only the incomplete-coverage models move (§5.1). Answering
107 is part of the score: the task is fixed-coverage forecasting, not selective prediction. Angles alone are
108 scored, since wrapping gives answers and failures one bounded scale that unbounded velocities lack.

109 **Verification record.** For each evaluation cell we retain the trajectory identifier, model and de-
110 ployment name, prompt, raw response, parser outcome, prediction, prompt and completion token
111 counts, and score. The released configuration fixes the data-generation solver, held-out seed, horizons,
112 prompts, timeout, and temperature. Two standards apply here. Local methods are reproducible:
113 their checkpoints and seeds regenerate the predictions exactly. Hosted models are not, because
114 a deployment can change or be withdrawn between evaluations, so for them the record is what

115 remains. We therefore release the per-cell responses rather than summary scores alone, which keeps
116 the reported API evaluation auditable after the deployments it measured are gone.

117 **Predictability horizon.** For chaotic systems, pointwise error past the Lyapunov time saturates to a
118 system-dependent constant for every model, so long-horizon error rankings carry little information.
119 We define the predictability horizon as the longest horizon a model tracks contiguously below a
120 threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that
121 exceeds τ . The contiguous rule is used because, after divergence, a later horizon can fall below τ by
122 angle-wrap coincidence. We sweep τ .

123 **Disclosure gap.** We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$. Positive values indicate a penalty
124 for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty,
125 we also run a matched experiment (§5.2) that toggles disclosure on identical trajectories.

126 4 Methods

127 **LLMs.** We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through
128 an OpenAI-compatible API (hosted deployments, June 2026), with a structured prompt (system
129 description, disclosed constants where applicable, initial state) and a JSON answer schema (Ap-
130 pendix A). Each cell uses one temperature-0 call with a uniform token and timeout budget; no
131 best-of- N aggregation is used. The LLM evaluation (Parts A and B, 3,960 cells) records prompt
132 and completion token counts and has an estimated nominal API cost of \$110.38 under the pricing
133 assumptions in the released analysis script.

134 **Time-series foundation models.** We run Chronos-2 (model card `amazon/chronos-2`; the Chronos
135 family is introduced by Ansari et al., 2024) locally on a single GPU in its standard history-conditioned
136 form: the 10 s pre- $t=0$ context window of all $2k$ state channels, resampled to at most 512 points,
137 forecasts to the target horizon (prediction length ≤ 64 steps). The comparison uses `chronos-2`
138 in univariate and multivariate modes; the latter forecasts all $2k$ channels jointly. The paired mode
139 comparison uses 20 trajectories per (k, regime) cell.

140 **Learned dynamics.** Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and
141 integrate at inference with `scipy`. We compare two objectives: derivative matching (MSE on finite-
142 difference $\dot{s}$) and rollout loss (MSE on the state after w Euler steps, with gradients through the unroll).
143 We also use mixed-window rollout, which trains on several window lengths jointly (Neural ODE:
144 $w \in \{10, 50\}$; HNN: $w \in \{5, 20\}$). The LNN’s loss requires second-order automatic differentiation;
145 the LNN trains at full scale (67k samples, hidden 256) on a CUDA GPU. The primary models are
146 trained on nine trajectories, one for each (k, regime) cell (Appendix B). A nested budget study
147 trains fresh Neural ODE and HNN checkpoints with $\{1, 5, 20\}$ trajectories per eligible cell and
148 seeds $\{101, 102, 103\}$. The Neural ODE uses all nine cells; because damping violates Hamiltonian
149 conservation, the HNN budget study uses only the three undamped normal-regime cells. Every
150 checkpoint is evaluated on the same two held-out seeds.

151 **Integrators.** Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations
152 of motion; their error is from discretization alone.

153 **Baselines.** Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity
154 ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast from the
155 matching (k, regime) training trajectory. The nearest-neighbor baseline uses the same data as the
156 learned models.

157 On both held-out seeds (equal-weighted; §5.3) we also fit SINDy with a degree-two polynomial library
158 and STLSQ, and nonlinear vector autoregression (NVAR) with two delays, degree-two features, and
159 ridge regression. Divergent rollouts are retained as failures and receive the same $\pi/2$ penalty.

160 **Lorenz replication.** A secondary Lorenz–63 test compares numerical references, persistence,
161 SINDy, NVAR, a Neural ODE trained on 20 single-seed trajectories, and zero-shot Chronos-2 on
162 five five-trajectory held-out seeds. We report training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s

Table 3: Out-of-sample cross-family comparison on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at $\pi/2$; 95% bootstrap CI. Inputs differ by family (Table 2); the ranking is conditional on them, this budget, and this system. RK4 generates the data and is included as a numerical reference. Configurations first evaluated in this work are marked $\dagger$; simple baselines are marked $\ddagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	98%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor $\ddagger$	0.725	[0.657, 0.797]	100%	baseline
7	linearized $\ddagger$	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence $\ddagger$	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed $\dagger$	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed $\dagger$	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity $\ddagger$	1.319	[1.241, 1.396]	100%	baseline
18	lnn $\dagger$	1.356	[1.282, 1.430]	94%	learned

(full configuration in Appendix C). HNN and LNN are omitted: Lorenz–63 is dissipative and non-Hamiltonian.

5 Results

5.1 Failure accounting and held-out generalization

The shared held-out grid contains 360 cells: 180 trajectories at horizons $\{1, 10\}$ s. Every model is evaluated on that grid. The Answered column in Table 3 gives the fraction of cells with valid predictions. Confidence intervals are 95% percentile bootstrap intervals from $B=10^4$ resamples of whole held-out trajectories (both horizon cells together), paired across models on a common resample; they are marginal per model, not simultaneous over the 19 rows, and for LLMs exclude per-call sampling variation.

Table 3 ranks the models on the pendulum under the access conditions in §3.3. The ranking is conditional on those inputs, this training budget, and this system; Section 5.4 tests whether it transfers. The numerical integrators sit at the top as references (RK4 0.059 at a finer step, symplectic 0.205, Euler 0.405), and every learned-dynamics model lands closer to the $\pi/2$ random baseline than to the RK4 and symplectic references. The lowest-error black-box entry is kimi (0.538 rad, 98% answered), whose interval is disjoint from the strongest baseline’s (nearest-neighbor, 0.725). That is a statement about one recorded June 2026 deployment under these evaluation conditions, not about model families; the Lorenz test rejects the family reading.

Under the equalized hidden-regime run, nearest-neighbor lookup (0.725) beats every learned model with a disjoint interval; the linearized pendulum (0.855) and persistence (1.004) sit above every learned model but HNN-rollout (0.972) on point estimates, with marginal intervals that overlap near the boundary. The stable learned models score 0.09–0.63 rad on their training trajectories against 0.97–1.14 held out (Table 4), a $1.7\text{--}11\times$ gap widest for the in-sample winner, the mixed-window Neural ODE.

Two further data-driven methods appear only on the two-seed replication subset, so we report them in Table 5 rather than in the main comparison, and their scores must be read with their coverage: SINDy answers 61% of cells and scores 0.883 rad on that subset, and NVAR answers 46% and scores 1.248. Because unanswered cells take the $\pi/2$ penalty, a majority of NVAR’s score and a substantial part of SINDy’s is imputation rather than forecast error, so these entries rank rollout stability together with accuracy.

Effect of missing predictions. grok answers 82% of the grid and lnn 94%: their ranks fall when missing cells receive $\pi/2$. On answered cells alone grok scores 0.729 rather than 0.881, ahead of every learned model; fixed coverage keeps its abstentions in the score. Under the strict π bound the LNN stays last and grok falls eight places, while full-coverage models are unchanged. The order among full-coverage models is invariant to the imputation constant; the rank of an incomplete-coverage model is not, which is why coverage enters the score. A prompting change shows the same mechanism: chain-of-thought leaves the ordering unchanged but is worse for the best LLM (+0.121 rad, CI [+0.061, +0.183]) because its unanswered cells rise from 6 to 50, moving the score through coverage, not accuracy (Appendix A).

5.2 Matched disclosure and identifiability

On the shared held-out set the hidden regime has lower error than the disclosed regime (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524). This looks favorable to withholding constants, but the regimes use different trajectories, and the hidden ones are slower and less divergent, so the comparison cannot speak to parameter recovery. When we instead toggle disclosure on identical trajectories for the three LLMs (900 paired cells on a separate matched grid; Table 8), the pooled sign changes: hiding the constants raises error by a small amount ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$, six tests). This pooled figure averages heterogeneous per-model effects (+0.211, +0.038, -0.042) and is carried by $k=1$ (+0.297; $k=2$ and $k=3$ reverse), so it bounds any penalty as small rather than estimating one. The unadjusted hidden-regime advantage is therefore due to trajectory difficulty, not disclosure.

The disclosure test also has a structural limit. The angular equations are unchanged by $(m, d) \mapsto (am, ad)$ and by $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable from an angle trajectory; absolute g , L , m , and d cannot be scored as independently recoverable. Asked to name the constants from a single state, the LLMs mostly return Earth-standard defaults, which is evidence of a prior, not of recovery (transcripts in the released records). A released check verifies both invariances to below 10^{-13} .

5.3 Replication across seeds and training budgets

We reran the local pendulum subset on an independent held-out seed with five trajectories per (k, regime) cell; the original seed has 20. Table 5 reports each seed separately and gives their unweighted mean. Three facts are seed-invariant: the numerical references have the lowest errors, Chronos-2 precedes the learned models, and HNN-rollout has the lowest error among the selected learned models. Two adjacent pairs invert between seeds—the two Chronos-2 modes, and SINDy versus HNN-rollout—so the ordering is stable at the family level but not between near-tied entries. Two seeds do not estimate seed-level uncertainty. The LNN’s low coverage was a hardware artifact, not model behavior: its inference solves a Hessian at each step, which exhausted the 8 GB GPU on the longest rollouts and returned those cells as failures. Re-running the LNN on CPU is the same evaluation with the same forecasts; it only recovers the dropped cells, raising primary-seed coverage from 66% to 94%. We use this CPU evaluation for the LNN on both seeds (the second answers all 90 cells, equal-seed rate 97%), so its numbers are produced identically and are comparable across seeds. The cells that still fail are genuine autograd-ODE divergences, scored at $\pi/2$ like any other failure, and the mean stays high at 1.366 rad.

Training-budget check. With three training initializations and the two held-out seeds, Neural ODE error decreases from 1.090 at one trajectory per cell to 0.968 at five and 0.861 at twenty (Table 10). HNN error on the normal regime falls from 1.269 to 0.974 and then remains 0.981. The HNN and Neural ODE rows cover different regimes and are not an architecture comparison. More data lowers Neural ODE error monotonically in point estimate (the $1 \rightarrow 5$ step is within the intervals, the $1 \rightarrow 20$ change is not); the HNN gain plateaus by 15 trajectories, and neither closes the gap to known-equation integration.

5.4 Cross-system test: Lorenz–63

We test whether the pendulum conclusions transfer to Lorenz–63, a chaotic system with a polynomial vector field. They do not. Both the identifiability result and the model ordering change.

The identifiability verdict flips. On the pendulum the physical constants are not recoverable from an angle trajectory (§5.2); on Lorenz, SINDy recovers the governing equations themselves, not merely accurate forecasts. The refit selects exactly the true sparsity pattern and returns $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ to coefficient error 2.2×10^{-13} (Appendix C); forecast error stays below 2×10^{-5} in every seed. The difference is the setting, not the audit: Lorenz-63 lies inside SINDy’s hypothesis class and is observed in full state, so its parameters are identifiable, whereas the pendulum’s absolute constants are not identifiable from angles under any method. Identifiability is a property of the observation model and the hypothesis class, and the check makes it explicit before recovery is scored.

The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25 trajectories), a Lorenz Neural ODE reaches normalized MAE 0.014 ± 0.008 , 0.079 ± 0.030 , and 0.277 ± 0.063 at horizons 0.1, 0.5, and 1.0 s (mean $\pm$ standard error over seeds), below both Chronos-2 modes at every horizon (multivariate 0.101 ± 0.018 , 0.691 ± 0.129 , 0.771 ± 0.097 ; univariate 0.126 ± 0.027 , 0.700 ± 0.102 , 0.732 ± 0.085). NVAR is close at 0.1 s (0.012) but drifts to 0.429 by 1.0 s, a persistence baseline trails (0.900 at 1 s). Every method answered all Lorenz cells (Table 5), so no failure imputation applies here and the pendulum’s $\pi/2$ constant, specific to the wrapped-angle scale, plays no role. This is the reverse of the pendulum, where the trained models lost to trivial baselines: here a system-trained model beats the zero-shot foundation model. As on the pendulum the families see different inputs (SINDy the library, the Neural ODE target-system training, Chronos a history window), so this is not an architecture race; the cross-system result rejects the pendulum ranking as a general ordering of model families.

The reversal does not rest on a favorable held-out seed (the training seed is fixed; §7). The Neural ODE beats both Chronos-2 modes in every one of the 15 seed-by-horizon comparisons, and SINDy stays below 1.7×10^{-5} in every cell. The identifiability half of the contrast is not a sampling claim: the pendulum symmetries are exact and verified to below 10^{-13} , and Lorenz-63 lies in SINDy’s library by construction.

The Lorenz test includes the local methods for which the comparison is well defined. It excludes the API LLM grid and the Hamiltonian and Lagrangian networks; a Hamiltonian model does not match the dissipative Lorenz flow. Those omissions limit what the test can establish, but not what it can refute: the pendulum result is only interesting if it generalizes, and one system on which the ordering and the identifiability verdict both invert is enough to withdraw that generalization. We claim no more than that, and in particular the Lorenz ordering is not offered as the correct one.

5.5 Real data: the reference model becomes the imperfect one

On real hardware the RK4 integrator is no longer ground truth but the imperfect model. We use an open video-tracked double pendulum [Myers et al., 2020] (three free-drop trials, 500 Hz, published point-mass constants, $\sim 0.35^\circ$ median tracking noise); Trial 1 serves only for calibration (nearest-neighbor database, SINDy fit, one-parameter damping fit), and Trials 2–3 supply 120 held-out states scored as before at $\{0.1, 0.5, 1.0\}$ s (Table 15, Appendix F).

RK4 with the published constants leads at 0.1 s (0.084 rad, about $14\times$ the tracking noise) but reaches 1.25 rad by 1.0 s, indistinguishable from persistence (1.23; paired difference $+0.02$, CI $[-0.12, +0.16]$); unmodeled friction and the point-mass simplification erase its advantage within a second. The fitted damping helps at one horizon and hurts at another (0.89 vs. 0.82 at 0.5 s, 1.15 vs. 1.25 at 1.0 s; the 1.0 s gain is significant, CI $[-0.202, -0.003]$). SINDy, exact on Lorenz, sits at baseline level here (1.54); the hardware’s dynamics are not in its library. Chronos-2 answers 89% of cells, since states shortly after the drop lack the 5 s context used here (Appendix F); failure accounting scores them at $\pi/2$. Three of the five checks run unchanged here—held-out generalization, failure accounting, and cross-system transfer, since the ordering differs from both synthetic orderings; the matched-disclosure and identifiability checks are not run on this apparatus, which has no LLM grid and no disclosure toggle.

5.6 Diagnostics

Behavior by horizon. Out-of-sample error by horizon appears in Table 13 (Appendix E). At 1 s the ordering follows the main comparison: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.86–1.25. At 10 s all models are in the chaotic regime; kimi (0.619) stays the lowest black-box method and the learned models reach 1.09–1.46,

but within-family ranks shift (deepseek falls from 0.996 to 0.909 while the Neural ODE rises from 0.892 to 1.328), so horizon-specific ranks are not read from the pooled score. Figure 1 shows the growth curves: the numerical integrators stay low through 1 s while every black-box family climbs toward the chaotic noise floor, and the $\{1, 10\}$ s window is where the families remain separable.

Breakdown by k and regime. Tables 6 and 7 (Appendix D) report error by link count and regime. Error rises with k for every model but the mixed-window Neural ODE. The coarse ordering (numerical below black-box below learned) holds at $k=2, 3$ only: at $k=1$ euler (0.271) falls among the Chronos modes and deepseek (0.872) drops into the learned band, so Table 3 averages over k and should not be read as a within- k ranking. The changed_hidden regime is easiest for most models; equalizing access (withholding the hidden constants, supplying normal-regime priors) raises learned-model hidden errors to 0.71–1.19 but does not remove it: the low-gravity constants ($g=1.62$) make the motion slower and less divergent, so the residual advantage reflects difficulty, and the regime remains easiest even with access equalized (§5.2). Resampling cells independently rather than whole trajectories changes no interval by more than ~ 0.03 rad, so within-trajectory dependence does not drive the widths.

Training-objective check. Table 11 reports the effect of the loss with architecture fixed. Under the equalized hidden-regime run, rollout loss is not uniformly better: mixed-window rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under our implementation the LNN is numerically unstable and generalizes poorly out of sample.

Physical-fidelity check. Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped regime, where true energy is constant (Table 12, drift over 10 s normalized by $E^*=g \sum_i m_i L_i$). The learned models do not match numerical integrators: the lowest-drift learned model is 0.779 against 0.066 for the symplectic integrator. Rolling the HNN out with a symplectic integrator changes its drift only from 2.065 to 2.036, so the integrator does not account for it; the fault is the learned Hamiltonian. The LNN’s 4.038 is over the 27% of bounded rollouts and is not comparable to the fully-bounded rows.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable (0.543 versus 0.549; no effect survives Holm correction), so the small per- k multivariate advantage in Table 6 is not evidence of joint-channel benefit.

6 Discussion

More data improves the Neural ODE from 1.090 to 0.861 rad without closing the gap to known-equation integration, and HNN improvement plateaus. Performance therefore depends on system, observations, hypothesis class, data, and access, none of which a scalar leaderboard encodes.

7 Limitations

The full LLM grid uses one held-out seed (the replication adds a second with fewer trajectories per cell); the budget study uses three initializations and two seeds; the Lorenz test excludes API LLMs, HNN, and LNN; the hardware check has three trials of one apparatus with filtered velocities; SINDy needs the true polynomial degree. Predictions are single endpoint states, conservation tests cover only the undamped regime, and we cannot rule out pendulum-like trajectories in the foundation models’ corpora. Intervals are marginal, not simultaneous, and incomplete-coverage scores are conditional on the $\pi/2$ imputation.

8 Conclusion

Generalization, failure accounting, matched disclosure, identifiability, and cross-system transfer each change a claim the raw leaderboard supports, so benchmarks should report them alongside training budgets and information access. Exact ground truth isolates each change to the check itself.

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A Prompt templates

The prompts used for every LLM evaluation cell are reproduced below verbatim.

390 **System prompt (direct / no-CoT).**

391 You are a careful physicist predicting the future state of a k -pendulum (a chain of
392 k point masses on massless rods, hinged in a planar chain, swinging under gravity
393 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
394 no code fences. Schema:
395 {"theta": [...k floats...], "omega": [...k floats...]}
396 theta_i is in radians (link i measured from straight-down,
397 positive counter-clockwise)
398 and omega_i is angular velocity in rad/s.

399 **System prompt (chain-of-thought).** Same preamble, then: "Reason step-by-step about the dynamics,
400 then output your final answer. Wrap the final answer in <answer>...</answer> tags containing ONLY a
401 single JSON object: {"theta": [...k floats...], "omega": [...k floats...]}."

402 **User prompt (coordinate modality).**

403 Number of links $k = \{k\}$
404 Physical constants:
405 g (gravity, m/s^2): $\{g\}$
406 L (rod lengths, m): $\{L\}$
407 m (point masses, kg): $\{m\}$
408 damping (viscous coefficient on ω): $\{\text{damping}\}$
409 Initial state at $t=0$:
410 θ_0 (rad): $\{\theta_0\}$
411 ω_0 (rad/s): $\{\omega_0\}$
412 Predict the state at $t = \{T\}$ seconds.
413 Return θ and ω as length- $\{k\}$ JSON arrays.

414 **Hidden-constants regime.** The constants block above is replaced by:

415 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
416 masses, and damping are NOT disclosed. You must infer dynamics from the initial
417 conditions alone.

418 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
419 <answer>...</answer> tags; the first $\{\dots\}$ object is JSON-parsed and validated to contain length- k θ and
420 ω arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

421 **Prompting, modality, and repeatability axes.** The released grid also varies prompting and input
422 modality; from the saved records (trajectory-level paired bootstrap throughout): (i) The chain-of-thought result is
423 in §5.1; for the best-scoring LLM it costs 0.659 vs. 0.538 rad through coverage, while for the other two models it
424 is indistinguishable (-0.038 , CI $[-0.102, +0.025]$; -0.041 , CI $[-0.107, +0.021]$). (ii) On the one-trajectory-
425 per-cell modality grid, replacing coordinates with a rendered image never helps (significantly worse for one
426 model, $+0.210$, CI $[+0.033, +0.392]$), adding an image to coordinates changes nothing measurable, and the
427 model without vision support fails every image cell, which fixed-coverage scoring records at $\pi/2$: modality is
428 an access variable, not a free axis. (iii) A second independent temperature-0 run of one model's 216-cell grid
429 agrees bit-identically on 172/216 cells but shifts the aggregate by $+0.073$ rad (CI $[+0.031, +0.120]$), so hosted
430 temperature-0 inference is not exactly repeatable; this measured drift is why hosted results are reproducible only
431 from the saved records.

432 **B Training and compute**

433 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate,
434 and train on a single 8 GB consumer GPU. Mixed- k rollout adds negligible parameters; its cost is wall-clock
435 time (the HNN mixed-window run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H
436 via autograd). Chronos-2 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and
437 three regimes, subsampled at $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the sample count, the training
438 set contains only *nine* trajectories, one for each (k, regime) cell. The budget study uses nested subsets of 1,
439 5, and 20 trajectories per eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or
440 180 trajectories; HNN sees 3, 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds
441 throughout.

Table 4: In-sample versus held-out error (rad) for the six stable learned models: mean over the nine training trajectories versus the shared held-out grid, both at horizons $\{1, 10\}$ s with the harness inference path. The in-sample best is the most overfit.

Model	In-sample	Held-out	Ratio
neural-ode	0.625	1.110	1.8
neural-ode-rollout	0.382	1.121	2.9
neural-ode-rollout-mixed	0.092	1.036	11.3
hnn	0.568	1.138	2.0
hnn-rollout	0.560	0.972	1.7
hnn-rollout-mixed	0.522	1.066	2.0

442 Compute cost. The API LLM evaluation is the only billed run (\$110.38 for 3,960 forecasts, June 2026
 443 deployments); local methods cost only compute.

444 Reproducibility records. The anonymized artifact accompanying this submission contains the code,
 445 prompts, trained weights, held-out specifications, and the per-cell records for the main held-out comparison
 446 (2,160 records), the matched disclosure test (1,800), the modality and prompting grid, and the Lorenz sweep;
 447 the two large sweeps, the training-budget and balanced-seed studies, are packaged as separate archives. All
 448 dataset and held-out seeds are fixed and recorded in the released configuration. LLMs are sampled once at
 449 temperature 0. The reported hosted-model numbers are recomputed from the saved records; re-calling the API is
 450 a new evaluation rather than a reproduction. Invalid cells are scored at $\pi/2$.

451 C Seed and system replication

452 The Lorenz-63 test uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t=0.01$ s, a 5 s context, one fixed
 453 training seed, and five fixed held-out seeds (all recorded in the released configuration). Each held-out seed
 454 contains five trajectories, giving 25 held-out trajectories in total. The Neural ODE is a two-hidden-layer, width-
 455 128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a degree-two polynomial
 456 library with STLSQ, fit to analytic derivatives at the sampled training states; the refit selects the true sparsity
 457 pattern exactly and recovers $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ with maximum absolute coefficient error 2.2×10^{-13} .
 458 NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in
 459 univariate and multivariate modes.

460 D Breakdowns by link count and regime

461 Reliability-adjusted mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s), by number of links
 462 (Table 6) and by regime (Table 7); models in the order of Table 3. Cells a model never produced (e.g. the LNN's
 463 diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 3.

464 E Additional result tables

465 This appendix gives full per-model values for the horizon breakdown (§5.6) and the predictability horizon
 466 (§3.4, Table 14), together with the regime definitions, the matched disclosure gap, and the training-budget,
 467 training-objective, and physical-fidelity studies. The corresponding summary statistics appear in the main text.

468 F Real-data check details

469 The hardware dataset [Myers et al., 2020] is openly released via Mendeley Data (CC BY 4.0 listing; the deposit
 470 includes a GPLv3 license file) with video tracking of both link angles at 500 Hz and per-frame uncertainty
 471 estimates (median 0.20° upper link, 0.43° lower). We use the published simplified point-mass parameters
 472 ($L = [0.172, 0.143]$ m, $m = [0.311, 0.111]$ kg, $g = 9.81$ m/s²). Angular velocities come from a Savitzky-
 473 Golay filter (window 42 ms, cubic, analytic derivative). The recordings begin with the pendulum positioned
 474 by hand, so the free drop is detected as the first time either link exceeds 4 rad/s; evaluation states are drawn
 475 evenly from 0.2 to 24 s after the drop, 60 per trial. The tracker zeroes its angle at the first frame, leaving a
 476 gravity-referenced offset per link; every trial ends hanging at rest with tracked angles at exact multiples of 360°
 477 (within 0.08°), which fixes the offset at zero independently of any model. A 20 ms physics-tracking check agrees,
 478 with error of 0.005–0.007 rad against a 0.006 rad tracking floor. A grid search over $\{0, 5 \times 10^{-4}, \dots, 2 \times 10^{-2}\}$
 479 selects damping 10^{-3} on Trial 1 at the 0.5 s horizon. SINDy uses a degree-two polynomial plus first-order

Table 5: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and their seed mean; horizons $\{1, 10\}$ s. The primary seed has 20 trajectories per (k, regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz-63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below 2×10^{-5} in every cell. Lower is better.

Pendulum model	Primary seed	Second seed	Seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	$<10^{-5}$	$<10^{-5}$	$<2 \times 10^{-5}$
neural-ode	0.014 (0.008)	0.079 (0.030)	0.277 (0.063)
nvar	0.012 (0.001)	0.262 (0.050)	0.429 (0.084)
chronos2-uni	0.126 (0.027)	0.700 (0.102)	0.732 (0.085)
chronos2-multi	0.101 (0.018)	0.691 (0.129)	0.771 (0.097)
euler	0.028 (0.005)	0.216 (0.023)	0.885 (0.120)
persistence	0.524 (0.054)	1.028 (0.123)	0.900 (0.057)

Table 6: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.440	1.556

Table 7: Out-of-sample error by regime. Learned-model hidden cells use the equalized hidden-regime run. The hidden regime remains easier for most models because its low-gravity constants give slower, less-divergent motion. The unadjusted regime contrast in §5.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.525	1.416	1.127

Table 8: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected. This experiment uses its own matched grid (3 models $\times$ 2 horizons $\times$ 150 trajectories = 900 pairs), not the shared held-out set, so its absolute errors are not comparable to those in Table 7. Per-model Δ values are computed before rounding.

Model	Disclosed	Hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

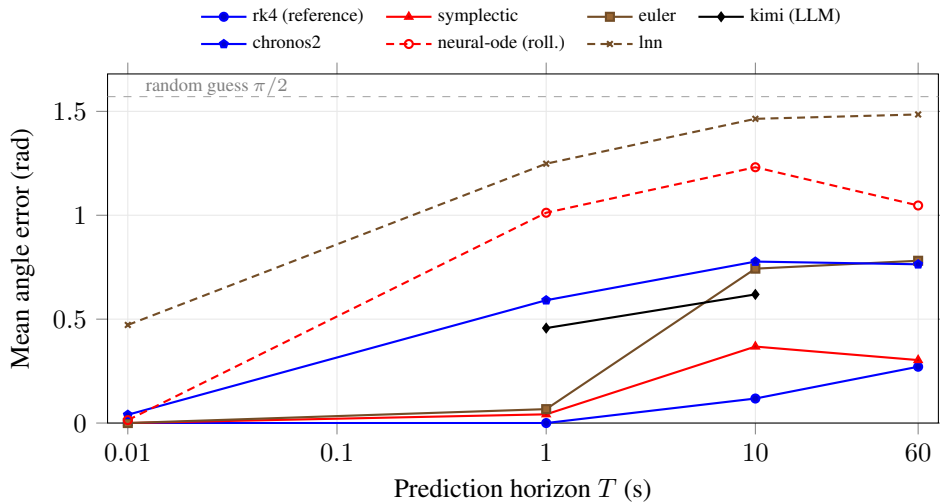


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs are evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 9: Physical regimes. The changed_hidden regime withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
changed_disclosed	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
changed_hidden	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 10: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals come from hierarchical resampling over initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Total trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

Table 11: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed-window	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed-window	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

Table 12: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^*=g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 13: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn	1.248	1.464

Table 14: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better; “—” means not even the first horizon stays below τ .

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01

480 Fourier library on Trial 1 with Savitzky–Golay derivatives; Chronos-2 receives a 5 s context resampled to 50 Hz
 481 and forecasts at most 50 steps. Failures are scored at $\pi/2$; Chronos-2’s are input-infeasible cells (insufficient
 482 context) rather than divergences, and the fixed-coverage rule scores both alike. The tracking dataset has been
 483 public since 2020 and may appear in foundation-model corpora; contamination would inflate Chronos-2’s score,
 484 so its baseline-level result is, if anything, conservative. Paired bootstrap differences over the 120 states at
 485 1.0 s: RK4 minus persistence $+0.018$ (95% CI $[-0.121, +0.155]$); damped minus undamped RK4 -0.103
 486 ($[-0.202, -0.003]$), so the damping gain at the longest horizon is significant while the RK4–persistence gap
 487 is not. States sampled along a trajectory are temporally dependent, so we repeat both with a moving-block
 488 bootstrap within each trial (blocks of 10 consecutive states, ~ 4 s): the conclusions are unchanged (damped
 489 minus RK4 $[-0.179, -0.019]$; RK4 minus persistence $[-0.131, +0.064]$), and remain so for block lengths
 490 from 2 to 12 s.

Table 15: Real-data check: mean wrapped angle error (rad) over both links on Trials 2–3 (120 held-out states), failures at $\pi/2$. The physics rows use the published constants; “damped” adds the Trial-1-fitted viscous coefficient. Tracking noise floor ~ 0.006 rad.

Model	0.1 s	0.5 s	1.0 s	Answered
physics-rk4	0.084	0.820	1.248	100%
physics-rk4-damped	0.086	0.892	1.145	100%
nearest-neighbor (cross-trial)	0.371	0.849	1.117	100%
chronos2	0.611	1.160	1.223	89%
constant-velocity	0.345	1.402	1.576	100%
sindy-real	0.490	1.501	1.535	99%
persistence	0.934	1.654	1.230	100%